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Nonlinearity in intergenerational income transmission: A cross-country analysis



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ABSTRACT

The aim of this paper is to explore nonlinearity in intergenerational income transmission. We use a set of occupational tables in different countries to test nonlinearity. We also empirically address how policy variables can affect nonlinearity. Our findings suggest that concavity is supported in those societies with less credit constraints, but with more poverty and income inequality; education has an increasing effect on convexity.

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1. Introduction

The seminal Becker and Tomes (1979, 1986) model is indisputably the main theoretical framework used by economists to understand the determinants of intergenerational income transmission. As Goldberger (1989) argued, they suggest a microeconomic analysis of family behavior drawing economists into areas formerly reserved to anthropologists, demographers and sociologists, and they also provide a rational choice interpretation to the mechanical model of intergenerational transmission developed by Galton (1886).

Using the theoretical framework of the Becker–Tomes (BT henceforth) model, a rich literature has focused on how to measure intergenerational income transmission, how to compare it across countries and over time.¹ A large set of empirical studies investigates the determinants of intergenerational income elasticity, that measures the degree to which a society gives equal opportunity of success to all its members, irrespective of their family background. If we consider that a higher intergenerational income elasticity implies a lower degree of social mobility, we have that the individuals from the lowest class have more difficult to move upward in the social class they were born (Solon, 1992; Zimmerman, 1992). The intergenerational income elasticity seems to vary with differences in factors relating to financial markets, education, human and social capital (Blanden et al., 2007; Bratsberg et al., 2007; Cardak et al., 2013; Chetty et al., 2014).

In the last decade, much attention has been given to *nonlinearity* in intergenerational income transmission. Couch and Lillard (2004) and Mazumder (2005) find a concave relationship for the US. Using quantile regression estimates and Canadian

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¹ See Solon (2002) and Black and Devereux (2010) for survey analysis.

data, [Grawe \(2004\)](#) finds that convexity occurs at the top of the income distribution; while a S-shaped relationship occurs at the bottom of the income distribution. [Bratsberg et al. \(2007\)](#) find that the pattern of intergenerational income persistence is linear for the US and the UK, and convex for the Nordic European countries.

Comparatively, as less attention has been given to the determinants of *nonlinearity* in intergenerational income transmission, the aim of this paper is to empirically investigate this neglected aspect in literature. Estimation methods, variable definitions and sample selection rules often differ widely across the previous empirical studies, including data from different samples, and comparability of results may be tricky. Thus, we use a set of occupational mobility tables provided in [Ganzeboom et al. \(1989a\)](#), which allows a comprehensive cross-country comparative analysis. This set of mobility tables has also the distinctive advantage of employing a consistent and well defined classification of socio-economic status. In details, we first present a simplified version of the BT model and the assumptions on the parameters that may yield nonlinearity; second, we explain the computational method to derive a test of nonlinearity in income transmission from the occupational mobility tables; then we estimate the determinants correlated with nonlinearity. Our main findings show that concavity occurs in less rich countries and with higher income inequality; convexity is related to those countries with more credit constraints, but with redistributive policies aimed to increase the human capital.

2. The Becker–Tomes model

As the BT model is very well known, we recall here only its essential elements using the approach in [Solon \(2004\)](#):

- the parent derives utility from their own lifetime consumption, C_{t-1} , and their child lifetime income, y_t

$$U_{t-1} = (1 - \alpha) \log(C_{t-1}) + \alpha \log(y_t) \quad (1)$$

where $0 < \alpha < 1$ denotes a coefficient of parent's altruism;

- the parent allocates his income y_{t-1} between his consumption, C_{t-1} , and investment in child's human capital, I_{t-1}

$$y_{t-1} = C_{t-1} + I_{t-1} \quad (2)$$

- the technology translating parental investment into child's human capital, h_t , is

$$h_t = \theta I_{t-1} + e_t \quad (3)$$

where $\theta > 0$ is a parameter indicating the efficacy of investments in the child's human capital, and e_t denotes child's initial endowment, influenced by nature and nurture, but orthogonal to I_{t-1} ;

- the child's income, y_t , depends on human capital

$$y_t = \mu + p h_t \quad (4)$$

where μ is market luck and p is the rate of return on human capital;

- the child's initial endowment in Eq. (3), e_t , evolves through family generations according to a AR(1) process

$$e_t = \delta + \lambda e_{t-1} + v_t \quad (5)$$

where $0 < \lambda < 1$ denotes the heritability coefficient and v_t is a stochastic error term.

Maximizing parent's utility of Eq. (1) under the budget constraint of Eq. (2) and using Eqs. (3)–(4), gives the optimal level of parent's investment in child's human capital

$$I_{t-1}^* = \alpha y_{t-1} - \frac{1 - \alpha}{p\theta} (\mu + p e_t). \quad (6)$$

Substituting Eq. (6) in Eqs. (3) and (4) we obtain the intergenerational income transmission equation

$$y_t = m + b y_{t-1} + \alpha p e_t \quad (7)$$

with $m = \alpha \mu$ and $b = \alpha p \theta$.

Since e_t follows the AR(1) process, as defined in Eq. (5), the OLS estimate of the slope coefficient β in the linear regression of y_t on y_{t-1} , that is the intergenerational income elasticity, is

$$\beta = \frac{b + \lambda}{1 + b\lambda} = \frac{\alpha p \theta + \lambda}{1 + \alpha p \theta \lambda}. \quad (8)$$

Eq. (8) helps understanding estimated β 's across countries and times in terms of the underlying structural parameters. In particular, the intergenerational income elasticity β is an increasing function of the heritability coefficient λ , the altruism coefficient α , the productivity of the educational system θ , and the return to human capital p .

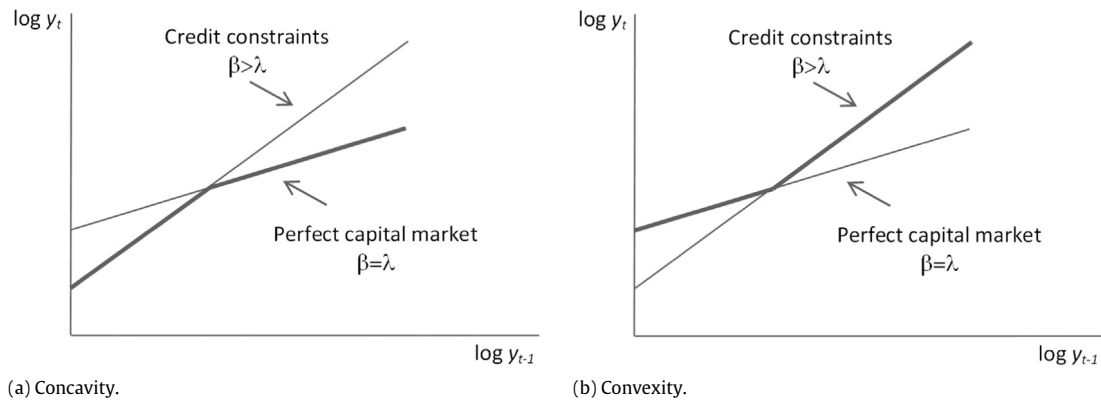


Fig. 1. Nonlinearity in intergenerational income transmission.
Source: Bratsberg et al. (2007).

2.1. Nonlinearity

Originally, Becker and Tomes (1986) conjectured the hypothesis of concavity in the relationship between child and parent income (Fig. 1(a)). Concavity is related to the imperfect capital market, it occurs if $\beta > \lambda$ and the borrowing constraints are more likely to impact on poor parents; if $\beta = \lambda$, then the perfect capital market occurs and even the poorest parents could borrow and make optimal investments in human capital. Recently, Bratsberg et al. (2007) conjecture the hypothesis of convexity (Fig. 1(b)), that occurs if the educational system is meritocratic, that is, the access to education is characterized by equal opportunity. This implies that $\theta = 0$ and, hence, $\beta = \lambda$ for the poorest families. Alternatively, if no family can borrow against child's future income, and we move away from Cobb–Douglas preferences admitting corner solutions, poor parents may not invest in their child's human capital (specially in presence of extensive public education programs), and so $\beta = \lambda$ for the poorest families and $\beta > \lambda$ for the richest families, obtaining convexity in the intergenerational income relationship.

In general, using the theoretical model discussed above, any heterogeneity of the parameters for different income groups implies a nonlinear relationship. If we assume that the heritability coefficient λ does not systematically differ across income groups, the slope of β may differ whenever $b = \alpha p \theta$ varies across families: any systematic relation between fathers' income and α , p and θ will produce nonlinearities in the income relationship. Just as an example, if the marginal productivity of human capital investment θ is decreasing (increasing) in father's income, θ will be higher for the poorest parents, generating a concave (convex). Thus, nonlinearity is clearly related to the assumptions on the parameters in Eq. (8).

3. An empirical investigation

As argued in the introduction, in literature we can find several empirical studies aimed to test nonlinearity in intergenerational income transmission for a range of different countries. However, comparability of results of these studies may be tricky, because estimation methods, variables definition and sample selection rules often differ widely across studies. Thus, in this section, we provide an empirical analysis of nonlinearity using a data set that has been constructed in such a way that cross-country differences are meaningful and do not derive from differences in data construction across countries. Furthermore, we explore the policy variables correlated with nonlinearity.

3.1. Occupational mobility tables

Our sample to test nonlinearity includes 149 occupational mobility tables, for 35 countries and at different years, presented in Ganzeboom et al. (1989a). This sample is a comprehensive and well structured data set, that allows a substantial degree of comparability among the different tables.

Ganzeboom et al. (1989a) present the cross-classification of fathers occupation by sons current occupation for representative national samples of men aged 21–64, with the characteristic that the tables conform to a well specified six category schema as follows (in descending order of socio-economic status): (1) Large proprietors, higher and lower professionals and managers; (2) Routine non-manual workers; (3) Small proprietors with and without employees; (4) Lower grade technicians, manual supervisors and skilled manual workers; (5) Unskilled and semiskilled manual workers; (6) Self employed farmers and (unskilled) agricultural workers.

This six class schema is a modification of the ten class schema, elaborated by Erikson et al. (1979), that define the so-called EGP class schema on the basis of similarity of the different groups in the labor market and work place in terms of four job attributes: sector, employment, skill level and supervisor status.

The reduction of the class schema with respect to the original EGP schema has five advantages (Ganzeboom et al., 1989a). First the aggregations pertain to categories that are small in size and their use would result in very sparse tables. Second, all the aggregations are of neighbors (or near neighbors) on the mobility dimension estimated with a scaled association model by Ganzeboom et al. (1989b) and Luijkx and Ganzeboom (1989). Third, the aggregations reduce the problem of missing information regarding the number of employees for some categories. Fourth, the aggregation of farmers with the agricultural workers evades the problems of non-comparability of self-employed farmers in Western countries with cooperative farm workers in the socialist world. Finally, this latter aggregation ignores the downward mobility that may occur if the sons of farmers work as agricultural or unpaid family workers.

3.2. A nonlinearity test

The mobility tables report occupational status, rather than wages or income, and as occupational class is an ordinal variable, it does not allow a direct testing for nonlinearity in intergenerational income transmission. To circumvent this problem we make some assumptions on the structure of the occupational classes. We first interpret income in the theoretical model as *lifetime* (or *permanent*) income, because occupational class can actually be considered a more accurate proxy than current income, which is mostly used in empirical applications. Second, we assume that individuals in higher occupational class earn (on average) more than individuals in lower occupational class (Wright, 1979; Marks et al., 1989).

In details, we aggregate the original six sons' occupational classes into a binary variable (say Y_s), which takes value 1 if the son is in any of the first three social classes. Under this assumption, son's expected status is simply given by the probability that $Y_s = 1$. Next, we aggregate fathers' social classes into three categories, and we let Y_f denote a discrete random variable, which takes three possible values: *high* when father belongs to social class 1 or 2; *middle* when father belongs to social class 3 or 4; and *low* when father belongs to social class 5 or 6.

Under the assumption that the mobility process is *stochastically monotone*, the expected status of sons coming from a high class father will be greater than the expected status of sons coming from middle class father, which in turn will be greater than the expected status of sons coming from a low class father. This assumption is actually tested in Dardanoni et al. (2012) using this set of occupational mobility tables.

Now, it follows that intergenerational income transmission will be:

$$\begin{aligned} \text{Concave : } & \frac{E[Y_s | Y_f = \text{middle}] - E[Y_s | Y_f = \text{low}]}{Y_{f,\text{middle}} - Y_{f,\text{low}}} > \frac{E[Y_s | Y_f = \text{high}] - E[Y_s | Y_f = \text{middle}]}{Y_{f,\text{high}} - Y_{f,\text{middle}}} \\ \text{Linear : } & \frac{E[Y_s | Y_f = \text{middle}] - E[Y_s | Y_f = \text{low}]}{Y_{f,\text{middle}} - Y_{f,\text{low}}} = \frac{E[Y_s | Y_f = \text{high}] - E[Y_s | Y_f = \text{middle}]}{Y_{f,\text{high}} - Y_{f,\text{middle}}} \\ \text{Convex : } & \frac{E[Y_s | Y_f = \text{middle}] - E[Y_s | Y_f = \text{low}]}{Y_{f,\text{middle}} - Y_{f,\text{low}}} < \frac{E[Y_s | Y_f = \text{high}] - E[Y_s | Y_f = \text{middle}]}{Y_{f,\text{high}} - Y_{f,\text{middle}}}. \end{aligned}$$

To calculate $Y_{f,\text{low}}$, $Y_{f,\text{middle}}$, $Y_{f,\text{high}}$ for all countries in the data set, we use the World Income Inequality Database (WIID), compiled by UNU-WIDER (2005), that contains a comprehensive collection of within country inequality data from several thousand underlying statistical surveys and is a very broad collection of national income distributions data. We use the income distribution estimate of the year which is closest to the period when the parents in the sample of occupational mobility table were making decisions about investments in their children's human capital. We use the deciles of the fathers' income distribution jointly with the marginal distribution of fathers' occupational class to get an estimate of $Y_{f,j}$ ($j = \text{low}, \text{middle}, \text{high}$).

We calculate

$$\mu = \frac{E[Y_s | Y_f = \text{middle}] - E[Y_s | Y_f = \text{low}]}{Y_{f,\text{middle}} - Y_{f,\text{low}}} - \frac{E[Y_s | Y_f = \text{high}] - E[Y_s | Y_f = \text{middle}]}{Y_{f,\text{high}} - Y_{f,\text{middle}}} \quad (9)$$

and the standard deviation of μ from a random sample of 1000 multinomial draws for each mobility tables. We finally estimate the standardized score $\hat{z} = \frac{\hat{\mu}}{\hat{\sigma}}$ for each table, which is asymptotically normally distributed. Using the 95% upper and lower critical values of the standard normal, we can infer that the relationship is linear when $\hat{z}$ lies inside the critical values, concave when $\hat{z}$ lies above the upper critical value and convex when it lies below the lower critical value. The standardized score $\hat{z}$ can be interpreted as a measure of nonlinearity in intergenerational income transmission.

Our sample consists of 63 observations. The lack of a larger sample size is caused by a lack of compatible income distribution and mobility tables. Table 1 summarizes our findings from the nonlinearity test. The linearity hypothesis in intergenerational income transmission cannot be rejected for 28 tables out of 63, intergenerational income transmission is convex for 8 tables and concave for the remaining tables. A glance at the table suggests that, similarly to Bratsberg et al. (2007), Nordic countries in general tend to have a convex relationship, the same is true for eastern bloc countries, in developing countries intergenerational income transmission seems to be concave.

Table 1
Nonlinearity in intergenerational income transmission.*

Country	Year	Patterns	$\hat{z}$	Country	Year	Patterns	$\hat{z}$
Australia	1987	Concave	2.92	Israel	1974	Concave	4.47
Canada	1973	Concave	2.26	Italy	1968	Concave	5.00
	1982w	Concave	2.35		1972	Linear	1.30
England	1967t	Linear	−0.88		1974	Concave	4.93
	1969	Linear	−1.45		1975p	Linear	0.47
	1972	Linear	−0.38	Japan	1975	Concave	5.32
	1974	Convex	−1.79	Netherlands	1967t	Concave	3.50
	1974p	Linear	1.33		1970	Linear	1.43
	1983	Linear	−0.84		1971	Concave	2.92
	1986	Linear	−0.71		1971e	Concave	5.49
Finland	1972s	Linear	−1.58		1974p	Concave	3.19
	1975p	Linear	−0.60		1976	Concave	3.73
	1980	Convex	−1.77		1977	Concave	1.96
	1982w	Concave	3.32		1977x	Concave	2.98
Germany	1959	Concave	2.13		1979p	Linear	−0.84
	1969	Concave	2.12		1982	Concave	3.09
	1969k	Concave	4.25		1982u	Concave	3.01
	1975p	Linear	0.60		1985	Concave	2.61
	1976z	Concave	4.02	Norway	1982w	Linear	−0.74
	1977z	Linear	0.81	New Zealand	1976	Concave	2.75
	1978	Linear	−0.39	Poland	1982	Linear	−1.48
	1978z	Concave	4.35		1987	Convex	−1.78
	1979z	Convex	−4.25	Scotland	1974	Linear	0.99
	1980p	Linear	1.56		1975	Convex	−2.09
	1980	Linear	0.51	Sweden	1972s	Linear	−1.05
	1980z	Linear	0.47		1973	Linear	−0.79
	1980a	Concave	2.41		1983w	Convex	−2.28
	1982a	Linear	0.36	USA	1983g	Linear	0.56
	1984a	Linear	0.32		1984g	Linear	0.71
Hungary	1982	Convex	−2.34		1985g	Linear	−1.10
	1986	Convex	−2.79		1986g	Concave	1.76
India	1971n	Concave	8.50				

* Critical values at the 5% significance level.

3.3. Regression analysis

We estimate the relationship between nonlinearity in intergenerational income transmission and a set of policy variables within the previous sample of 63 observations. The empirical specification is constructed as follows. We regress the standardized score, $\hat{z}$, on a constant, the GDP per capita (transformed in natural logarithmic terms), the Gini index (as a measure of income inequality), the private credit by deposit money banks and other financial institutions divided by GDP (as a measure of credit accessibility), the public education expenditure per capita (transformed in natural logarithmic terms), and the average schooling years, which is a common measure of human capital used in the empirical literature. The [Tables A.1](#) and [A.2](#) in the [Appendix](#) report, respectively, the description with the data sources of the variables, and summary statistics.

[Table 2](#) in column (1) presents the ordinary least-squares (OLS) regression results. The coefficient upon income inequality is positive and the coefficient upon the GDP per capita is negative, which suggest that poorer and more unequal societies tend to have a more concave relationship in intergenerational income transmission. In other words, in poor countries and when income inequality is high, the prospects of the sons of middle class families are closer to those of poor families, while when incomes are higher and more equally distributed, the prospects of the sons of middle class families are closer to those of rich ones. The coefficient upon the credit accessibility is positive, which suggests that countries with fewer credit constraints tend to have a more concave pattern. For the variables related to education, we find that the coefficient upon the public education expenditure per capita is positive, while the coefficient upon the human capital is negative, suggesting that convexity is more related to the schooling level rather than to the amount of education expenditure and, hence, public investment devoted to higher education level may imply a decreasing effect on concavity.

3.4. Robustness check

This section investigates the robustness of the previous results through three tests: (i) substitution of the policy variables with other proxies; (ii) alternative occupational class schema (iii) endogeneity analysis.

Substitution of the policy variables

Since one may be concerned that the findings was driven by the specific set of policy variables, four regressions have been run, that include substitution of the policy variables by other proxies. In details, income inequality, measured by the

Table 2

OLS estimates (dependent variable: z-score, strength of concavity).

	(1)	(2)	(3)	(4)	(5)
Constant	35.043 (3.275)***	26.265 (1.946)*	15.551 (2.492)**	23.169 (2.544)**	33.979 (3.396)***
ln GDP per capita	–7.125 (3.264)***	–3.615 (1.416)	–2.757 (3.671)***	–4.658 (2.403)**	–8.189 (3.982)***
Income inequality	0.263 (5.350)***		0.248 (5.476)***	0.264 (6.658)***	0.363 (5.556)***
ln public education expenditure per capita	3.772 (2.280)**	1.082 (0.582)		2.778 (2.068)**	5.368 (3.433)***
Human capital	–0.128 (0.710)	–0.169 (0.787)	–0.157 (0.992)		–0.277 (1.503)
Credit accessibility	2.407 (2.029)**	3.267 (2.335)**	2.719 (2.276)**	2.315 (2.139)**	
Income interdecile ratio		0.104 (3.552)***			
Public education expenditure (% of GDP)			0.686 (2.586)**		
Primary school attained (% of population aged 25 and over)				–0.092 (5.000)***	
Secondary school attained (% of population aged 25 and over)				–0.014 (0.449)	
Higher school attained (% of population aged 25 and over)				–0.311 (3.951)***	
Lending interest rate					0.011 (0.89)
Number of observations	63	63	63	63	52
R ²	0.399	0.261	0.405	0.511	0.371
Adj R ²	0.346	0.196	0.453	0.449	0.294
F-test	7.56	4.03	7.76	8.22	4.84

Absolute *t*-statistics are displayed in parentheses under the coefficient estimates.

* Statistically significant at the 10% level.

** Statistically significant at the 5% level.

*** Statistically significant at the 1% level.

Gini index, is replaced with the income interdecile ratio, that has been calculated as the ratio of the highest income decile to that in the lowest decile; the public education expenditure per capita (measured in natural log) is replaced with the public education expenditure as share of GDP; the average number of schooling years with the education attainment levels (primary, secondary and higher); the private credit by deposit money banks and other financial institutions as share of GDP with the lending interest rate, that is the bank rate that usually meets the short and medium-term financing needs of the private sector. The [Tables A.1](#) and [A.2](#) in the [Appendix](#) report, respectively, the description with the data sources and summary statistics of these proxies. Each regression is run as anew with the new proxy. Using the OLS method, the results, reported in [Table 2](#), columns (2)–(5), show a strikingly consistent pattern. The coefficients confirm their sign and order of magnitude. Convexity occurs in richer countries with higher educational attainment; while concavity is associated with more income inequality and less private credit constraint.

Alternative occupational class schema

A second cause for concern can be that the standardized score $\hat{z}$ has been calculated using a social hierarchical approach of the class schema proposed in [Ganzeboom et al. \(1989a\)](#) and, hence, using a vertical dimension of the occupational class. One may be consequently worry about the sensitivity of our results to the specific calculation of the dependent variable, $\hat{z}$. We therefore use the class structure proposed in the Erikson–Goldthorpe (E–G) schema ([Erikson and Goldthorpe, 1992](#)). Class structure refers to the social position of the individuals as identified by the relations within the labor market. On the basis of this approach, the E–G schema collapses to the following three categories:

- *non manual workers* including: large proprietors, higher and lower professionals and managers; routine non-manual workers; small proprietors with and without employees;
- *farm workers* including: farmers and smallholders;
- *manual workers* including: lower grade technicians, manual supervisors and skilled manual workers; unskilled and semiskilled manual workers; agricultural workers.

Next, we collapse fathers' social classes into these three categories and Y_f (that is a discrete random variable) takes the values: *high* when father belongs to *non manual workers*; *middle* when father belongs to *manual workers*, excluding the agricultural workers; *low* when father belongs to *farm workers*, including the agricultural workers. Grouping farm with the agricultural workers finds reason in the aggregation of our sample from [Ganzeboom et al. \(1989a\)](#). The binary variable Y_s , related to the sons' social classes, takes value 1 if the son belongs to *non manual workers*.

Table 3

OLS estimates (dependent variable: z-score, strength of concavity), E–G schema.

Constant	12.939 (0.981)
ln GDP per capita	–1.868 (0.717)
Income inequality	0.165 (2.575)**
ln public education expenditure per capita	0.472 (0.244)
Human capital	–0.224 (0.141)
Credit accessibility	1.873 (1.219)
Number of observations	63
R ²	0.234
Adj R ²	0.167
F-test	3.48**

Absolute *t*-statistics are displayed in parentheses under the coefficient estimates.

** Statistically significant at the 5% level.

Table 4

2SLS estimates (dependent variable: z-score, strength of concavity).

	(1)	(2)	(3)	(4)
Constant	38.135 (3.331)***	60.922 (4.257)***	35.168 (3.265)***	48.33 (2.729)***
ln GDP per capita	–8.885 (3.686)***	–13.395 (4.369)***	–7.177 (3.290)***	–8.769 (2.848)***
Income inequality	0.409 (4.423)***	0.321 (5.404)***	0.263 (5.012)***	0.209 (3.121)***
ln public education expenditure per capita	5.299 (2.896)***	8.825 (3.735)***	3.844 (2.211)**	4.016 (2.128)**
Human capital	–0.186 (0.278)	–0.383 (2.004)**	–0.143 (0.641)	–0.036 (0.204)
Credit accessibility	1.529 (0.274)	3.123 (2.289)**	2.396 (1.919)*	4.959 (1.544)
Number of observations	63	63	63	60
R ²	0.323	0.299	0.399	0.342
Adj R ²	0.263	0.237	0.346	0.281
F-test	5.44	4.86	7.56	5.61
F-first stage	9.95	275.4	15.72	8.63
Wu–Hausman test (<i>p</i> -value)	0.05	0.00	0.81	0.41
Sargan test (<i>p</i> -value)	0.13	0.55	0.15	0.97
Endogeneity test for:	<i>Income inequality</i>	<i>Public education expenditure per capita</i>	<i>Human capital</i>	<i>Credit accessibility</i>
Instruments:	Income per capita growth rate;	Urbanization rate;	Percentage of “primary school attained” in the total population (male);	Income per capita growth rate;
	Population growth rate	School-aged population on the total population	Percentage of “primary school attained” in the total population (female);	Urbanization rate;
			Percentage of “secondary school attained” in the total population (male);	Inflation rate
			Percentage of “secondary school attained” in the total population (female);	
			Percentage of “higher school attained” in the total population (male);	
			Percentage of “higher school attained” in the total population (female)	

Absolute *t*-statistics are displayed in parentheses under the coefficient estimates.

* Statistically significant at the 10% level.

** Statistically significant at the 5% level.

*** Statistically significant at the 1% level.

Table A.1

Description of variables and data sources.

Variable	Description	Source
<i>Policy variables</i>		
ln GDP per capita	Natural Log of GDP per capita (US \$)	Maddison (2007)
Income inequality	Gini index	UNU-WIDER (2005)
Credit accessibility	Private credit by deposit money banks and other financial institutions divided by GDP	Beck et al. (1999)
ln public education expenditure per capita	Natural log of public education expenditure per capita (US \$)	World Bank (2012)
Human capital	Average number of years of education received by people ages 25 and older	Barro and Lee (1994)
<i>Alternative policy variables</i>		
Income interdecile ratio	Ratio of the highest income decile to that in the lowest	UNU-WIDER (2005)
Public education expenditure (% of GDP)	Public education expenditure as share of GDP	World Bank (2012)
Primary school attained (% of population aged 25 and over)	Percentage of “primary school attained” in the total population	Barro and Lee (1994)
Secondary school attained (% of population aged 25 and over)	Percentage of “secondary school attained” in the total population	Barro and Lee (1994)
Higher school attained (% of population aged 25 and over)	Percentage of “higher school attained” in the total population	Barro and Lee (1994)
Lending interest rate	Interest rate that meets the short and medium-term financing needs of the private sector	World Bank (2012)

Following the E–G schema, we calculate the standardized score $\hat{z}$ (as previously explained in Section 3.2) and the regression results are displayed in Table 3, which are qualitatively similar to those reported in Table 2, column (1). The coefficients confirm their sign for all the variables and order of magnitude for almost all the variables. Differently to the previous results, we find that the positive marginal effect of the public education expenditure per capita on concavity is reduced; if we consider this result together with the increase of the negative marginal effect of the human capital, it seems that we have more complementarity between the education public expenditure and schooling level in the E–G schema rather than in the class schema proposed in Ganzeboom et al. (1989a).

Endogeneity analysis

As the policy variables may be partially endogenous to the standardized score $\hat{z}$, the two-stage least squares (2SLS) approach has been employed to control for omitted variable bias, potential reverse causality and measurement error. We instrumented the policy variables as follows. The instruments for the income inequality are the income per capita growth rate and the population growth rate (Brenner et al., 1991). The urbanization ratio and the share of school-aged population on the total population have been selected as instruments for the public education expenditure (Akanbi and Schoeman, 2010; Busemeyer, 2007). Several studies have shown that parents' education is an important factor for the educational outcome of their children (Dearden, 1998; Gang and Zimmermann, 2000; Nielsen et al., 2001); however, as we do not have data on the parents' educational attainment, and in order to guarantee transmission of education from the father's generation to the son's generation, we instrumented the average number of years of schooling in the year t with the educational attainment (primary, secondary and higher) in the year $t - 20$; we also consider the educational attainment by sex (male, female). The instruments for the credit accessibility are income per capita growth rate, the urbanization ratio and the inflation rate (Beck et al., 2008). The Table A.3 in the Appendix reports, respectively, the description with the data sources and summary statistics of the instruments.

The 2SLS estimates are reported in Table 4 and they are both qualitatively and quantitatively similar to those obtained with OLS estimates. Namely, they confirm that concavity is mainly associated to lower GDP, fewer credit constraints, higher income inequality and lower schooling level. The Sargan test rejects the hypothesis that the residuals are correlated with instruments for all the policy variables, and, hence, the instruments are valid. Also the F -statistics in the first stage regressions show that the instruments are not weak. The Wu–Hausman test rejects the hypothesis that public education expenditure is exogenous and income inequality is endogenous at the 5% significance level. Therefore, the 2SLS estimates should be preferred to OLS estimates in these two cases.

4. Conclusions

The main aim of this paper has been to examine nonlinearity in intergenerational income transmission. Differently to existing empirical studies, we provide a cross-country analysis using a sample of 63 social mobility tables, which includes 17 countries in different years. We reject linearity in 35 tables, and find that Nordic and eastern bloc countries tend to have a convex pattern of intergenerational income transmission. Furthermore, we have proceeded to explore the determinants of nonlinearity. Our findings suggest that concavity is associated with more income inequality and less rich countries; convexity is related to those countries with more credit constraints and human capital. These findings appear to be robust to different proxies of the policy variables, occupational class schema and to the endogeneity analysis.

Table A.2
Summary statistics.

Variable	Mean	S.D.	Min	Max	N.Obs
<i>Dependent variable</i>					
Strength of concavity	1.2	2.5	−4.2	8.5	63
<i>Policy variables</i>					
ln GDP per capita	9.4	0.4	6.8	10.0	63
Income inequality	30.9	5.3	20.9	44.9	63
ln public education expenditure per capita	6.5	0.6	3.0	7.2	63
Human capital	7.6	2.1	1.2	12.1	63
Credit accessibility	0.6	0.3	0.0	1.2	63
<i>Alternative policy variables</i>					
Income interdecile ratio	11.0	8.1	3.8	33.0	63
Public education expenditure (% of GDP)	4.9	1.2	2.1	7.0	63
Primary school attained (% of population aged 25 and over)	38.4	15.5	3.8	66.3	63
Secondary school attained (% of population aged 25 and over)	15.3	11.6	0.1	39.5	63
Higher school attained (% of population aged 25 and over)	6.6	5.0	0.7	22.6	63
Lending interest rate	12.7	11.5	6.4	89.2	52

Table A.3
Description, data source and summary statistics of the instruments.

Instrument	Source	Mean	S.D.	Min	Max	N.Obs
Income per capita growth rate	Maddison (2007)	2.2	2.5	−4.2	7.7	63
Population growth rate	Maddison (2007)	0.6	0.7	−0.5	3.8	63
Urbanization rate	World Bank (2012)	70.4	9.1	20.0	86.2	63
School-aged population on the total population	World Bank (2012)	22.8	3.9	16.4	41.0	63
Inflation rate	World Bank (2012)	9.0	8.7	0.9	55.3	60
Percentage of “primary school attained” in the total population (male)	Barro and Lee (1994)	50.0	18.9	1.7	84.9	63
Percentage of “primary school attained” in the total population (female)	Barro and Lee (1994)	7.3	7.9	0.1	36.8	63
Percentage of “secondary school attained” in the total population (male)	Barro and Lee (1994)	1.4	1.9	0.1	8.5	63
Percentage of “secondary school attained” in the total population (female)	Barro and Lee (1994)	45.9	15.0	9.8	78.9	63
Percentage of “higher school attained” in the total population (male)	Barro and Lee (1994)	8.7	6.9	0.4	27.1	63
Percentage of “higher school attained” in the total population (female)	Barro and Lee (1994)	4.3	2.6	0.6	11.9	63

The policy implications are that concavity occurs in those countries with more inequality, both in terms of income and opportunity; convexity exists in those countries that provide support to public investment devoted to education and adopt public policies devoted to alleviate the persistence of poverty and income inequality.

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Appendix. Description of variables, data source and summary statistics

See [Tables A.1–A.3](#).

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